

A reproducible analysis

Change scores in a synthetic training study

Workshop participant

2026-07-29

1 Introduction

Literate programming keeps explanation and computation in the same source document (Knuth 1984). This reduces opportunities for reported values, figures, and tables to become detached from the analysis that produced them.

1.1 Research question

How did scores change between baseline and follow-up, and how did the descriptive change differ between the comparison and intervention groups?

The analysis aims to:

- calculate an individual change score;
- summarise change within each study group; and
- visualise the association between training hours and change.

For participant i , the change score is:

$$\Delta_i = y_{i,\text{followup}} - y_{i,\text{baseline}}.$$

Note

The data are entirely synthetic. They contain no real participants and cannot provide evidence about a real intervention.

2 Methods

The CSV is read from a relative path, and the change score is derived in code.

The dataset contains 40 fictional participants, with 20 in the comparison group and 20 in the intervention group.

3 Results

```
summary_table <- study |>
  group_by(group) |>
  summarise(
    participants = n(),
    mean_change = mean(change),
    sd_change = sd(change),
    .groups = "drop"
  )

knitr::kable(summary_table, digits = 1)
```

Table 1: Change scores by study group.

group	participants	mean_change	sd_change
comparison	20	3.0	1.2
intervention	20	10.9	2.7

Table 1 shows a larger average change in the intervention group than in the comparison group.

```
ggplot(study, aes(training_hours, change, colour = group, shape = group)) +
  geom_point(size = 2.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Training hours",
    y = "Change in score",
    colour = "Study group",
    shape = "Study group"
  ) +
  theme_minimal()
```

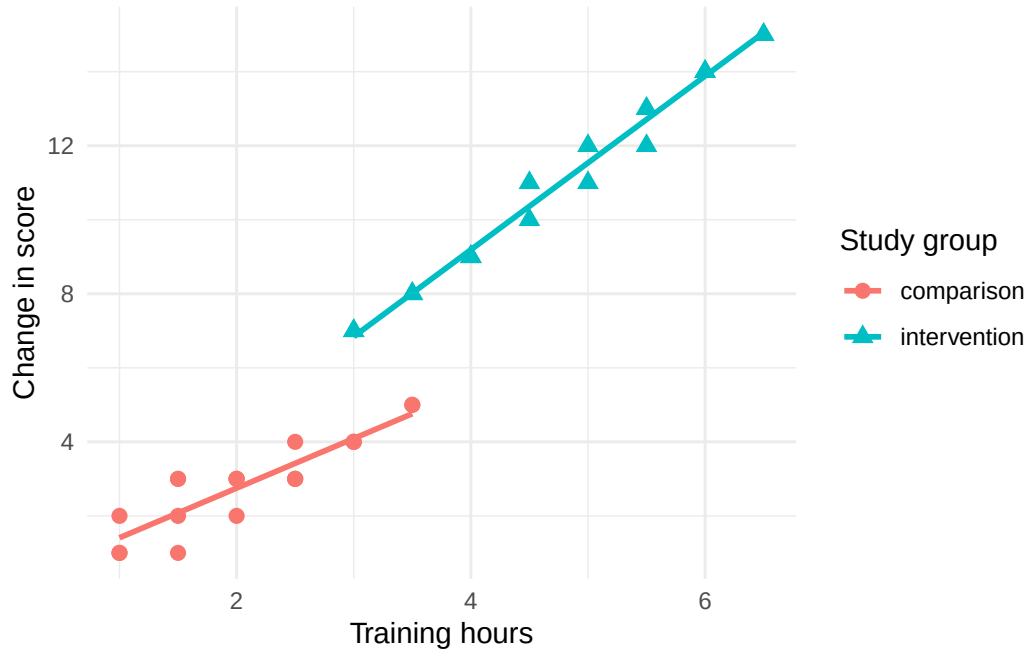


Figure 1: Change score by training hours and study group.

The constructed relationship in Figure 1 shows that group membership, training hours, and change are all associated in this synthetic dataset.

4 Discussion

The intervention group has the larger descriptive mean change. This comparison does not establish that the intervention caused the difference: the data are synthetic, participants were not randomised, and training hours differ systematically between groups. A real analysis would require a design-appropriate model, uncertainty estimates, and substantive assessment of possible confounding.

Reproducible computation keeps the numerical summary aligned with the source data, but it does not remove the author's responsibility for design, interpretation, or transparent limitations (Sandve et al. 2013).

5 Reproducibility

The source data, transformations, prose, and output instructions are stored in one project. The following information records the R environment used for this render.

sessionInfo()

R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK

locale:

[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Australia/Melbourne

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] ggplot2_4.0.3 dplyr_1.2.1 readr_2.2.0

loaded via a namespace (and not attached):

[1] Matrix_1.7-5	bit_4.6.0	gtable_0.3.6	jsonlite_2.0.0
[5] compiler_4.6.1	crayon_1.5.3	tidyselect_1.2.1	parallel_4.6.1
[9] splines_4.6.1	scales_1.4.0	yaml_2.3.12	fastmap_1.2.0
[13] lattice_0.22-9	R6_2.6.1	labeling_0.4.3	generics_0.1.4
[17] knitr_1.51	tibble_3.3.1	pillar_1.11.1	RColorBrewer_1.1-3
[21] tzdb_0.5.0	rlang_1.3.0	xfun_0.59	S7_0.2.2
[25] bit64_4.8.2	otel_0.2.0	cli_3.6.6	mgcv_1.9-4
[29] withr_3.0.3	magrittr_2.0.5	digest_0.6.39	grid_4.6.1
[33] vroom_1.7.1	hms_1.1.4	nlme_3.1-169	lifecycle_1.0.5
[37] vctrs_0.7.3	evaluate_1.0.5	glue_1.8.1	farver_2.1.2
[41] rmarkdown_2.31	tools_4.6.1	pkgconfig_2.0.3	htmltools_0.5.9

References

Knuth, Donald E. 1984. "Literate Programming." *The Computer Journal* 27 (2): 97–111.
<https://doi.org/10.1093/comjnl/27.2.97>.

Sandve, Geir Kjetil, Anton Nekrutenko, James Taylor, and Eivind Hovig. 2013. “Ten Simple Rules for Reproducible Computational Research.” *PLOS Computational Biology* 9 (10): e1003285. <https://doi.org/10.1371/journal.pcbi.1003285>.